Round 11 - One Mechanism Proposal

Mechanism: Response-State Closure (RSC)
Represent only the quantities consumed by the next-layer mean closure. Do not carry kappa3 sources or dense legs. Carry per layer four dense objects: C (covariance), Y21 (kappa3 iij slice), K4d (kappa4 diagonal), and an auxiliary response matrix R capturing regeneration of odd structure. The key approximation is that the missing deep content acts through a regenerated response field rather than through explicit source histories.

State

C: n x n. Y21: n x n. K4d: n. R: n x n. Total O(n^2) memory. R is updated recursively and replaces explicit source transport.

Update Rule

R(l+1)=(W■W)R(l)W^T + alpha*Y21(l)+beta*C(l). Variance, d3 and K4d are read out from linear combinations of C, Y21, K4d and R. No source identities are stored.

Cost

Transport of C ~2 units/layer. Y21 ~2 units/layer. R ~2 units/layer. Readout and diagonals <1 unit/layer. Total about 6-7 units/layer or roughly 96-112 units over 16 layers, below the 125-unit target.

Stage 1 Falsifier

Free-run from layer 1 using only stored teacher layer-1 state. Compare C off-diagonal, Y21, d3 and K4d at layers 7, 11, 14 and 15. Kill immediately if any channel exceeds 1.5x chain error at two deep layers or diverges faster with depth.

Teacher Check

Fit alpha and beta on layers 3 and 7. Evaluate frozen coefficients on layers 11, 14, 15 and 16. Required result: channel errors no worse than the chain and ideally improved on K4d and variance.

Stage 2 Target

Integrate into the chain. Plausible path: match channel accuracy while removing the 41-unit old tier and most join work. Target total cost about 102-120 units. To beat the chain score, MSE must remain within roughly 1.07x chain at 102 units or 0.95x chain at 115 units.

Specific Missing Term

The hypothesis is that the omitted information is not source memory itself but repeated ReLU regeneration of odd response structure, summarized by R. If Stage 1 succeeds but Stage 2 misses, enrich R with one extra matrix R2 driven by K31 and K22.

Why This One

All exact source carriers are closed by measurement. This proposal changes representation entirely: it tracks the response field that produces the channels consumed by means, not the underlying kappa3 tensor.